

Week 5, Lecture 1 WOD – Answer Key

Choosing a Data-Generating Model

NRES 746 – Instructor Answer Key (not for student distribution)

This WOD is ungraded, so this key is a quick reference for circulating during the 8-10 minute window, not a rubric. Nothing here has a single correct answer. Each data set was simulated from a specific generating model (the “truth”, below), but several other answers are defensible. For each part the key gives the truth, the acceptable variants, and the red flags to listen for. Whatever a student picks, look for four things:

1. The **function for the expected value** has a shape that matches the pattern in the figure (decline, saturation, S-shape) and respects the range of the response.
2. The **noise distribution** respects the type of data (discrete count, 0/1, positive continuous).
3. The **leftover noise parameter** is named (σ , k , a gamma shape, ...), or the student notices that the distribution has none (Bernoulli, Poisson, exponential).
4. The two lines are written as *expected value plus noise around it*, not as a model of the raw data.

The generating models behind the figure

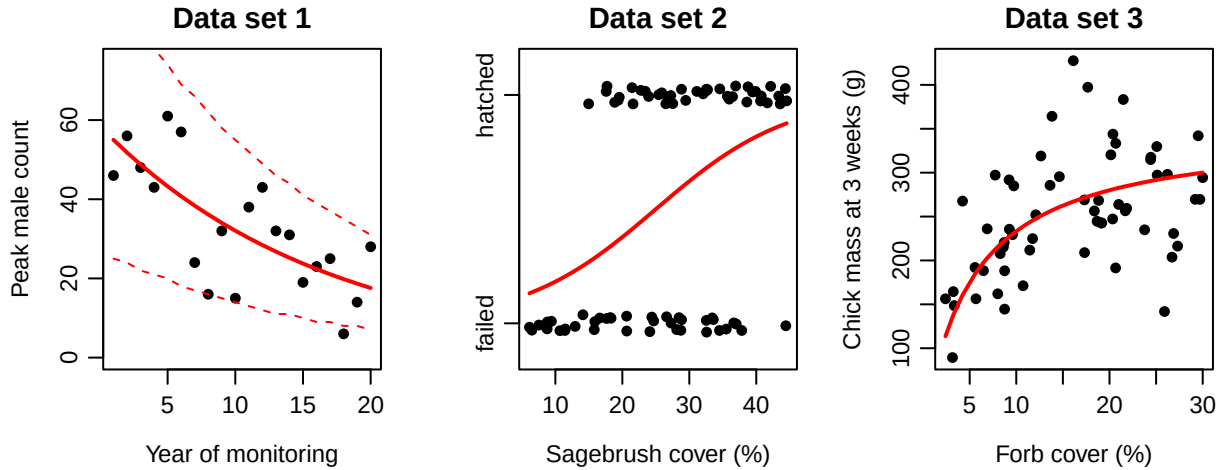
The chunk below reproduces the handout’s simulated data exactly (same seed and order of draws).

```
set.seed(746)
yr <- 0:19
lek_mu <- 55 * exp(-0.06 * yr)
lek <- rlnbinom(20, mu = lek_mu, size = 8)

cover <- runif(80, 5, 45)
p_true <- plogis(-2.5 + 0.1 * cover)
nest <- rbinom(80, 1, p_true)

forb <- runif(60, 2, 30)
mass_mu <- 350 * forb / (5 + forb)
mass <- rgamma(60, shape = 25, rate = 25 / mass_mu)
```

Data set	Expected value	Noise around it
1. Lek counts	$55 \exp(-0.06 t)$, $t = 0, \dots, 19$	Negative binomial, mean = expected value, size $k = 8$
2. Nest fate	$\text{logistic}(-2.5 + 0.1 \text{ cover})$	Bernoulli(p), $p = \text{expected value}$
3. Chick mass	$350 \text{ forb} / (5 + \text{forb})$	Gamma, mean = expected value, shape 25 (scale = mean / 25)
4. Hen movement	constant (no data plotted)	Described only; gamma, shape 0.8, mean 250 m is consistent



Red solid lines are the true expected values. The dashed red lines in panel 1 are the 5th and 95th percentiles of the true negative binomial. The x -axis in panel 1 is year of monitoring $= t + 1$, while the generating function uses $t = 0, \dots, 19$.

Part 1: Data set 1 – lek counts

Truth and best answer.

Expected value: negative exponential, $\mu_t = N_0 e^{-rt}$. Noise distribution: negative binomial with mean μ_t and size k .

$$\mu_t = N_0 e^{-rt} \quad Y_t \sim \text{NegBin}(\text{mean} = \mu_t, \text{size} = k)$$

The prompt says the counts fall by a constant *percentage* per year, which is exactly what an exponential function does ($\mu_{t+1}/\mu_t = e^{-r}$ for every t). The noise parameter is k : the variance is $\mu_t + \mu_t^2/k$, so smaller k means more spread.

1b. Why a negative binomial rather than a Poisson. Counts are discrete and non-negative, so Poisson and negative binomial are the natural first candidates. The prompt (and the figure) say the scatter is large. The Poisson variance equals the mean, so we can check how much scatter a Poisson would allow:

```
# Poisson would give Pearson dispersion (mean squared standardized residual) = 1
mean((lek - lek_mu)^2 / lek_mu)

## [1] 3.604747

# fraction of counts outside mu +/- 2 sqrt(mu) (about 5% under a Poisson)
mean(abs(lek - lek_mu) > 2 * sqrt(lek_mu))

## [1] 0.4

# noise SD at the start and end of the series: Poisson vs negative binomial
rbind(Poisson = sqrt(lek_mu[c(1, 20)]), NegBin = sqrt(lek_mu[c(1, 20)] + lek_mu[c(1, 20)]^2 / 8))

##           [,1]      [,2]
## Poisson  7.416198 4.194049
## NegBin   20.811655 7.501084
```

The dispersion is about 3.6 rather than 1, and 40% of the counts fall outside a range that should hold about 95% of Poisson counts. At the start of the series the Poisson would allow an SD of about 7 while the negative binomial (truth) has about 21.

Acceptable variants.

- *Poisson* is a reasonable first guess for counts. Treat it as a good answer if the student flags that the scatter looks too large for it (equidispersion, from Monday's lecture).
- *Normal*, *lognormal*, or *gamma* are defensible approximations for large counts (these run from 6 to 61), if the student says so. A normal with constant σ would not shrink its noise as the mean falls.
- Any decreasing function is fine as long as it can't go negative: a straight line $\mu_t = a - bt$ eventually crosses zero, which is the "doubt" for a linear function.

Red flags. Choosing Poisson without any comment on the spread; a Ricker, Michaelis-Menten or logistic function for a monotone decline in time (they don't have the right shape); leaving out k (a negative binomial is not complete without it).

Part 2: Data set 2 – nest fate

Truth and best answer.

Expected value: a logistic function of cover. Noise distribution: Bernoulli (a binomial with size 1).

$$p_i = \frac{1}{1 + e^{-(a+b \text{ cover}_i)}} \quad Y_i \sim \text{Bernoulli}(p_i)$$

There is no leftover noise parameter. For a 0/1 response the expected value *is* the probability of success, and the variance $p_i(1 - p_i)$ is fixed by that mean.

2b. Why Bernoulli. The response takes only two values, so the noise distribution must put all its probability on 0 and 1. A normal, gamma, or Poisson can't do that.

Why a bounded function. The prompt hints that a probability can't leave $[0, 1]$. A straight line is fine over the observed range but not beyond it:

```
lp <- lm(nest ~ cover)
range(fitted(lp))
```

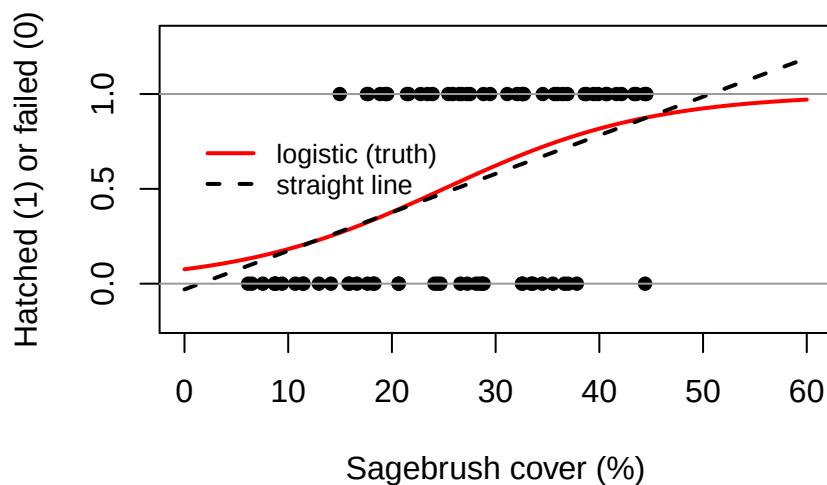
inside $[0, 1]$ over the observed 5-45%

```
## [1] 0.09444888 0.87484141
```

```
predict(lp, data.frame(cover = c(0, 60)))
```

outside $[0, 1]$ when extrapolated

```
##          1          2
## -0.03024553  1.18902509
```



Acceptable variants.

- *Michaelis-Menten with an asymptote of at most 1*, $p = a \text{ cover} / (b + \text{cover})$ with $a \leq 1$, is bounded, increasing, and defensible. A maximum-likelihood fit of both to these data (not something the students can do yet) finds the logistic slightly better, with negative log-likelihood 47.5 against 49.0, so the data don't strongly separate them.
- *Binomial*, if the student says each nest is a single trial (size 1). A binomial with $N = 80$ is a red flag: nests are 80 separate Bernoulli observations, not one experiment with 80 trials.

Red flags. Normal noise for a 0/1 response; a straight-line expected value with no comment on the bounds; applying the function to the 0/1 outcome itself instead of to the probability of success; adding a “ σ ” to a Bernoulli model.

Part 3: Data set 3 – chick mass

Truth and best answer.

Expected value: Michaelis-Menten. Noise distribution: gamma with mean μ_i and a shape parameter α as the noise parameter.

$$\mu_i = \frac{a \text{ forb}_i}{b + \text{forb}_i} \quad Y_i \sim \text{Gamma}(\text{mean} = \mu_i, \text{shape} = \alpha)$$

Here a is the asymptotic mass and b is the forb cover at which the expected mass reaches half of a . Mass is positive, and its spread grows with its mean (below), which is what a gamma with a fixed shape gives.

3b. Why gamma. The response is continuous and positive, which excludes Poisson and Bernoulli and makes a normal only an approximation. The spread grows with the mean:

```
bins <- cut(forb, c(0, 8, 16, 30))
res <- mass - mass_mu
tapply(mass, bins, mean)           # mean mass in each forb bin

##      (0,8]      (8,16]      (16,30]
## 189.6135 239.1104 278.6681

tapply(res, bins, sd)              # SD of the residuals rises with the mean

##      (0,8]      (8,16]      (16,30]
## 47.62183 49.38487 64.20530

tapply(res / mass_mu, bins, sd)    # CV roughly constant; the truth is 1 / sqrt(25) = 0.2

##      (0,8]      (8,16]      (16,30]
## 0.2933069 0.2072792 0.2281783
```

Acceptable variants.

- *Lognormal*: also positive with spread proportional to the mean. If the student puts the function on the *log* mean, that models the median rather than the mean: $E[Y] = e^{m+s^2/2}$. To keep the function on the expected value, use $m = \ln \mu_i - s^2/2$. With $\mu = 200$ and CV 0.2 that is $m = 5.279$, $s = 0.198$.
- *Normal with constant σ* : defensible here. The residual SD only rises from about 48 to 64 g across the range, and with mean masses of 190 to 280 g and a CV of 0.2, a normal rarely goes negative. Ask whether the spread grows with the mean, and whether they'd trust a normal outside the observed range.
- *Function*: any saturating shape in the grab bag is fine (Michaelis-Menten, a power law with exponent below 1, a logistic-type curve). A hump (Ricker) or a straight line doesn't match the description (“gains shrink at high cover”).

Worth raising. A Michaelis-Menten function passes through the origin, so it predicts zero mass at zero forb cover. Chicks are not born weighing nothing, so an intercept would be a reason to doubt it if the data extended to low cover.

3c (if time). For a gamma with shape α and scale s , the mean is αs . Setting this equal to μ_i gives

$$s = \frac{\mu_i}{\alpha}, \quad \text{Var}(Y_i) = \alpha s^2 = \frac{\mu_i^2}{\alpha}, \quad \text{CV} = \frac{1}{\sqrt{\alpha}}.$$

The shape therefore sets the *relative* noise: a large shape means small relative scatter, and the standard deviation grows in proportion to the mean. In R (rate = 1/scale):

```
alpha <- 25; mu <- 200
y <- rgamma(1e6, shape = alpha, scale = mu / alpha) # equivalently rate = alpha / mu
c(mean = mean(y), CV = sd(y) / mean(y), one_over_sqrt_alpha = 1 / sqrt(alpha))
```

```
##                mean                CV one_over_sqrt_alpha
##          200.0403487          0.1999501          0.2000000
```

Notation note. The handout writes the gamma shape as α so that it doesn't collide with the a students typically use for a function's parameter (the intercept, or the Michaelis-Menten asymptote). Bolker writes the gamma shape as a , so students who consult the book will see the other symbol. Accept any consistent naming.

Part 4: If time – a fourth data set

Best answer. The expected value is a **constant**, μ , because no predictor is available. The model still has an expected value, it just doesn't vary. The noise must be positive and right-skewed with a few long moves: an exponential, gamma, or lognormal distribution with mean μ .

$$\mu_i = \mu \quad Y_i \sim \text{Gamma}(\text{mean} = \mu, \text{shape} = \alpha)$$

All three candidates reproduce the description (“a typical hen moves 100-200 m, a few days more than a kilometer”) when the mean is about 250 m:

```
mu <- 250
cv <- 1.5; s_ln <- sqrt(log(1 + cv^2)); m_ln <- log(mu) - s_ln^2 / 2
data.frame(
  distribution = c("Exponential (CV = 1)", "Gamma, shape 0.8 (CV = 1.12)", "Lognormal, CV = 1.5"),
  median_m = c(qexp(0.5, 1 / mu), qgamma(0.5, 0.8, scale = mu / 0.8), exp(m_ln)),
  prob_over_1km = c(pexp(1000, 1 / mu, lower.tail = FALSE),
                    pgamma(1000, 0.8, scale = mu / 0.8, lower.tail = FALSE),
                    plnorm(1000, m_ln, s_ln, lower.tail = FALSE))
)
```

```
##                distribution median_m prob_over_1km
## 1      Exponential (CV = 1) 173.2868    0.01831564
## 2 Gamma, shape 0.8 (CV = 1.12) 156.6723    0.02642187
## 3      Lognormal, CV = 1.5 138.6750    0.03439894
```

The exponential has no noise parameter (its CV is fixed at 1). The gamma adds a shape and can make the tail heavier (CV above 1) or lighter. The lognormal has a log-scale SD as its noise parameter.

Red flags. A normal distribution (symmetric around 250 m, puts probability below zero, and cannot produce a few very long moves); a Poisson (distance is continuous); a uniform. “There is no predictor so there is no model” misses the point: an intercept-only model is still a model, with a constant expected value, and it is what you would fit before adding covariates such as habitat or time of year.

Discussion prompts for the neighbor comparison

Where students disagree, these are the productive questions.

- **Lek counts:** what would a Poisson predict for the scatter, and does the figure agree? (The answer is in 1b.)
- **Nest fate:** a straight line fits the observed range. What goes wrong if you use it to predict for a nest with 60% cover?
- **Chick mass:** gamma, lognormal, and normal are all defensible. What would you check in a residual plot to tell them apart?
- **Hen movement:** exponential, gamma, and lognormal all match the description. What extra data would separate them?